

# Prediction

You just watched a token’s final vector land in its neighborhood on the map. In Vector Space, that landing was about meaning, the model understanding the words you put in. Prediction is the flip side, the output: it takes the position at the end of your text and turns it into the next word.

## THE RANKED LIST

The model doesn’t just grab the single closest token. It compares that final position against **every** possible next token, scoring each by how closely the two line up on the map. Then it ranks them by probability: the best fit on top, and a long tail of less likely options below.

Then it takes one, almost always from the top, adds it to the text, and runs the whole thing again for the token after that. That pick has a name: **prediction**. Score every possible next token, choose one, then repeat.

## ONE DIAL ON THE PICK: TEMPERATURE

There’s a control on how boldly the model commits to the top of that ranked list: **temperature**.

Temperature is the variation dial. It changes how predictable or surprising the model is when choosing words. Low temperature makes answers more repeatable. High temperature makes them more varied, creative, and sometimes weird. The middle is the balanced default, best for everyday writing.

Temperature does not make the model smarter. It changes how much the model sticks to its most likely next choice.

Here’s that ranked list for “See you \_\_\_”. Watch how the same candidates redistribute as temperature rises. This is a simplified demo: real models use other sampling controls too, and exact numbers vary by model.

### How temperature reshapes the picks

|             |  Low |  Medium |  High |
|-------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| tomorrow    | <div><div></div></div> 100%                                                             | <div><div></div></div> 65%                                                                  | <div><div></div></div> 40%                                                                 |
| later       | <div><div></div></div> 0%                                                               | <div><div></div></div> 20%                                                                  | <div><div></div></div> 28%                                                                 |
| soon        | <div><div></div></div> 0%                                                               | <div><div></div></div> 10%                                                                  | <div><div></div></div> 17%                                                                 |
| there       | <div><div></div></div> 0%                                                               | <div><div></div></div> 3%                                                                   | <div><div></div></div> 7%                                                                  |
| again       | <div><div></div></div> 0%                                                               | <div><div></div></div> 1%                                                                   | <div><div></div></div> 4%                                                                  |
| other words | <div><div></div></div> 0%                                                               | <div><div></div></div> 1%                                                                   | <div><div></div></div> 4%                                                                  |

You probably won’t see this dial in regular ChatGPT, Claude, or Gemini. The app usually picks the temperature for you. But the idea still matters: you can ask for the effect instead, requesting *your best single answer* for consistency or *ten different options* for range.

## WHAT TEMPERATURE IS NOT

Low temperature doesn’t mean correct. It means repeatable. A low-temp model will give you the same answer every time, even when that answer is wrong. Temperature controls variation, not accuracy.

These odds didn’t come from nowhere. The model learned every one of them from billions of examples, long before you typed a word. That’s **training**.